



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER II, SESSION 2024/2025

BACHELOR OF COMPUTER SCIENCE (BIOINFORMATICS)

SECJ3553 ARTIFICIAL INTELLIGENCE - SECTION 01

PROJECT ASSIGNMENT 3

THEME: HEALTHCARE

(MINDCARE AI - INTELLIGENT MENTAL HEALTH SUPPORT CHATBOT)

GROUP MEMBERS	MATRIC NO
CHIN PEI WEN	A23CS0065
CHUA SHANG YEET	A23CS0297
KOO XUAN	A23CS0300
TAN ZHAO HONG	A23CS0188
TAY CHING XIAN	A23CS0307

LECTURER'S NAME : PROF. MADYA DR. ROLIANA IBRAHIM

SUBMISSION DATE : 12 JANUARY 2026

Table of content

1.0 Overview.....	3
2.0 Intelligent Agent.....	3
3.0 PEAS Model.....	4
4.0 PEAS Model Diagram.....	5
5.0 Explanation In Proof of Concept (POC).....	6
6.0 Behaviours Of Agent To Achieve The System Goals.....	7

1.0 Overview

As an intelligent agent, MindCare AI is based on the Knowledge Representation that was developed in Assignment 1 and the State Space Search model from Assignment 2, this assignment describes how the agent perceives inputs, reasons about mental health conditions, and takes appropriate actions. The PEAS model formally describes the behavior of MindCare AI and its performance measures, environment, sensors, and actuators. This assignment also explains how each component is implemented in the proposed proof of concept and the way the agent behaves within the system.

2.0 Intelligent Agent

MindCare AI is a medical aid agent of mental health, which accepts input of users, reasoned about the emotional state and generated appropriate responses. It processes user messages independently and tries to identify symptoms to determine risk of safety and make a decision which can be helpful or emergency. The agent will be developed to keep the users safe, as well as offer consistent and pertinent mental health services. This will see the MindCare AI perform well in a healthcare support set-up.

3.0 PEAS Model

Performances	Accuracy of symptom detection, Response time, Safety protocol effectiveness, User satisfaction
Environments	Online mental health support environment where users interact with the chatbot via a web or mobile interface.
Actuators	Screen display (text responses), Crisis hotline information, Breathing exercise guides, Coping recommendations, Mood tracking suggestions
Sensors	Text input analyzer (NLP), Symptom keyword detector, Sentiment analyzer, Risk level detector, Context tracker

MindCare AI is an AI chatbot that deals with mental health care, where text is analyzed to identify the emotional state and provide a user with the necessary recommendation. It targets the effective symptom identification, rapid response, and efficient safety measures to guarantee user safety and satisfaction.

The space is an online social platform where members support each other through web or mobile sharing their feelings of stress, anxiety or crisis-related ideas. The chatbot has to reply with compassion and security in this delicate environment.

Text-based actuators are empathetic responses, breathing exercises guidance, coping response, mood tracking response, and crisis hotline resources in the case of high-risk.

Actuators include text-based outputs such as empathetic responses, breathing exercise guides, coping strategies, mood tracking suggestions, and crisis hotline information for high-risk cases.

Sensors include Natural Language Processing (NLP) for text analysis, keyword detection, sentiment evaluation, risk assessment, and a context tracker to maintain coherent, relevant conversations. These enable MindCare AI to perceive user conditions accurately and deliver supportive interventions.

4.0 PEAS Model Diagram

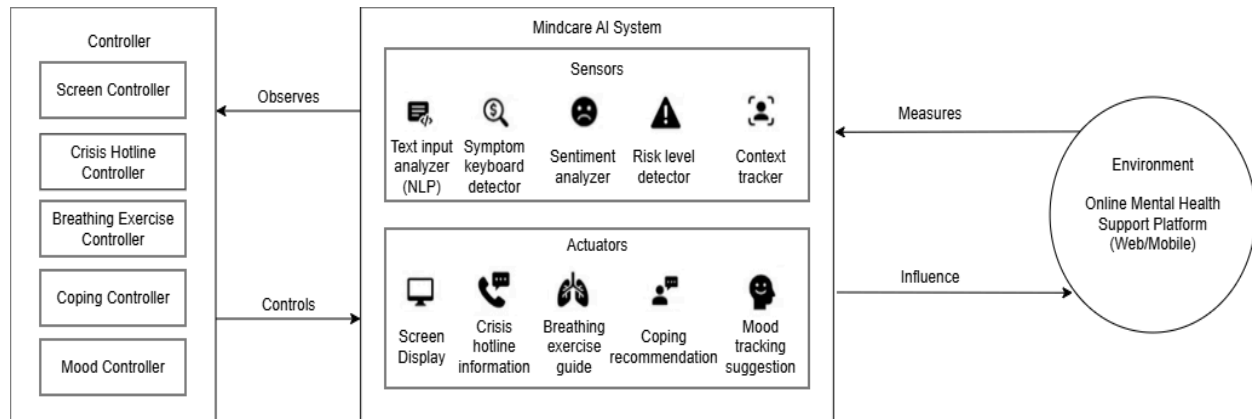


Figure 4.1: PEAS Model

Figure 4.1 depicts the architectural framework of the Mindcare AI System, modeled as a closed-loop control system. User interactions are recorded in the Environment (the Online Mental Health Support Platform), where the process starts. Sensors process these inputs and quantify user data using tools like risk level detectors and NLP analyzers. A central controller observes this data and chooses the best response tactics. Lastly, the Controller instructs Actuators—like breathing guides or crisis hotlines—to carry out particular actions that affect the Environment, giving the user individualized mental health support.

5.0 Explanation In Proof of Concept (POC)

Performance:

MindCare AI operates in the Proof of Concept (POC) focusing on proper detection of the emotional state, timely response and proper safety management. The system employs the Natural Language Processing (NLP) and sentiment analysis to recognize emotional patterns, like stress, anxiety, or emotional distress in user text input. The rapid response time has been used to ensure that the users get immediate response and safety measures are used to deal with more dangerous cases of emotional issues. The system will offer compassionate and applicable answers to increase user satisfaction and offer trustworthy mental health support.

Environment:

MindCare AI setting in the POC is an online mental health support program that is available on web or mobile applications. The system presents itself to users who input text messages that reflect their emotions, issues or mental health issues. This is a dynamic sensitive environment because the emotional states of the users can vary during interaction. The virtual environment enables the agent to keep perceiving the input of users and adjusting its responses to deliver relevant emotional support in a secure and regulated healthcare environment.

Actuators:

The MindCare AI POC actuators are text-based components that provide emotional support and guidance to the users. They consist of responsive, understanding reactions, suggestions of coping strategies, teaching breathing exercises, and advising of mood tracking. The system delivers safety-related information and urges users to find relevant external help in cases where the high emotional risk is identified. The agent can affect user well-being through these actuators to provide structured and supportive communication.

Sensors:

The MindCare AI POC sensors are virtual elements that interpret user input in real time. They consist of NLP text processing modules, sentiment analyzers to determine the emotional tone, mental health indicators key-word detectors, and risk level detectors to determine the emotional severity. Context tracker is a conversation history tracker that enables the agent to produce coherent and context-relevant responses. These sensors combined together allow the correct perception of the emotional condition of the user and informed decision-making by the agent.

6.0 Behaviours Of Agent To Achieve The System Goals

The MindCare AI project is focused on developing a smart mental health support chatbot that will act as an autonomous agent with the purpose of helping users to manage their emotional well-being. The fundamental purpose of the agent is to empathise with user emotions, give supportive advice and ensure user safety in a digital healthcare setting.

The system can be worked as an intelligent agent, which interacts with users using a chatbot environment. With the help of sensors, the agent receives user inputs constantly in the form of text messages and analyses emotional data such as sentiment, keywords, and the context of conversation. Those inputs allow the agent to analyze the user's current emotional state and give the relevant response strategy.

Once a user has begun a conversation, the input is run through the agent to identify the emotional indicators and assess the level of risk. For example, when a user is stressed or anxious, the agent will provide them with supportive messages and coping skills. In case the user changes his or her emotional state in the conversation, the agent adapts its responses accordingly in order to maintain relevance and support.

The agent has a constant feedback loop in which the context of the conversation is updated after each interaction by the agent. This allows MindCare AI to gain a better understanding of the emotional status of the users with time and offer more personalized support to the users. The system is constantly evaluating the efficiency of its responses according to user engagement and emotional dynamics.

Moreover, the agent is concerned about safety through the detection of the signs of increased emotional risk. When such trends are identified, the agent adjusts his or her behaviour so that emphasis is placed on reassuring calming guidance and responsible support recommendations. As a result, the system is aligned with its objective of offering safe, supportive, and reliable mental health support.

In general, MindCare AI behaves like a goal-oriented, reactive and context-aware and intelligent agent that is constantly perceiving, reasoning and acting with the aim of supporting user's mental well-being in the context of an online healthcare environment.